EPIC-Rec: Evidence-Preserving Counterfactual Explanations for Top-K Graph Recommendation

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Abstract

Counterfactual explanations for recommender systems name a small set of past interactions whose removal changes a recommendation. Existing methods remove the target item but leave the rest of the list to chance, and they do not ask whether the reasons behind the remaining recommendations survive the edit. We study the stricter question: can one recommendation be removed with a minimal edit to the user's history while every other top-K item stays in the list and keeps at least one of its original, score-contributing evidence paths? We formulate this as a constrained minimum-mask problem with three conditions—removal (C1), membership preservation (C2) and evidence preservation (C3)—and introduce EPIC-Rec (Evidence-Preserving Interaction Counterfactuals for Recommendation). On an inductive, additive path-scoring recommender the problem becomes a mixed-integer linear program that we solve for a margin variant, using lazy constraint generation. Every optimality or infeasibility claim is re-checked under a second configuration of the same solver, HiGHS. That cross-check is not a formal certificate. On MovieLens-1M, across 2,000 user–target pairs from 200 users, a mask that removes the target exists for 100.00% of pairs, a mask that also keeps the other $K-1$ items for 83.30% [80.92, 85.75], and one that also keeps their evidence for 73.10% [70.11, 76.05]. On Last.fm the same three rates are 95.70% [94.75, 96.60], 30.05% [28.05, 31.90] and 28.05% [26.15, 30.00]. A target-score order that stops at removal keeps the list and the evidence in only 27.5% of the MovieLens edits that do remove the target, and in 12.4% of such Last.fm edits. Those two rates are conditional on removal. On all 2,000 pairs the same MovieLens edits satisfy C1 + C2 + C3 for 21.80%. Surgical counterfactuals are common on MovieLens and uncommon on Last.fm. The Last.fm gap appears already when the list must be kept, before evidence is required.

1 Introduction

Counterfactual explanations justify an individual algorithmic decision by naming a small change to the input that would have produced a different output [27]. In recommendation, the input a user controls is their interaction history, and an explanation takes the form "had you not watched X and Y, you would not have been recommended Z" [10, 26]. Recent work has extended such explanations to graph-based recommenders [6, 7, 18], to learned and attribution-based explainers [3, 4, 11], to sequential models [16], and to evaluation protocols that look beyond the top-1 item [19, 21]. The closest work, CoSRec [32], learns sparse interaction masks for a fixed graph recommender, verifies them by recomputing the ranking, and reports how much of the surrounding list remains stable.

What these methods do not provide is a *guarantee* about the rest of the list. A mask that removes the target may also push other items out, and it may destroy the evidence that justified

them. Stability is measured after the fact, not enforced. For a user who wants one unwanted recommendation gone, or for an auditor who wants to know whether a recommendation can be explained independently of its neighbours, the relevant question is stricter:

> *Can we remove a specific recommendation with a minimal edit to the user's history while preserving every other top-K item and at least one of its original, score-contributing evidence paths?*

Why would one recommendation need to be removed without disturbing the others? Three situations motivate the question. First, user control: a user who presses "not interested" on one item—a title watched for a child, a one-off gift purchase, a product already bought elsewhere—expects that item to disappear, not the rest of a list they were satisfied with. Second, privacy: a single recommendation can reveal a sensitive inference about the user, and removing it should not degrade unrelated suggestions. Third, explanation and audit: even when nothing is actually removed, a minimal counterfactual ("had you not interacted with A and C, T would not have been recommended") is a verifiable account of why an item was recommended, and transparency obligations for recommender systems, such as those in the EU Digital Services Act [8], ask platforms to explain the main parameters of their recommendations and to offer users control over them. When surgical removal is impossible, that fact is itself informative: it shows that the recommendation is entangled with others in the list and cannot be explained or withdrawn in isolation.

We make four contributions.

1. A problem formulation with a joint guarantee: target removal (C1), exact top-K membership preservation for the other $K - 1$ items (C2), and survival of at least one original, score-contributing evidence path for each of them (C3), with an explicit ranking rule and an explicit evidence unit (Section 3).

2. An exact solution method. On an inductive, additive path scorer every score is linear in the binary mask, and a margin version of the problem becomes a mixed-integer linear program (MILP) that we solve with lazy constraint generation; C3 reduces to covering constraints (Section 4). We measure how often the margin version and the original problem differ.

3. A measured cost of each constraint on MovieLens-1M and Last.fm: how many user–target pairs admit a mask, how many are reported infeasible for P_ε on $D_u(t)$, and the paired increase in minimum mask size from C1 to C1 + C2 to C1 + C2 + C3, stratified by target rank and history length (Section 6).

4. Evidence that constraint-unaware heuristics rarely satisfy C2 and C3 by chance, and a comparison of heuristics and beam search with the exact optimum.

The counterfactual edge-mask mechanism builds on our earlier work on counterfactual inverse link prediction (CILP) [15], which removes edges from unipartite social graphs. EPIC-Rec moves this idea to user–item–entity graphs, replaces a single link prediction with a ranked list, and adds the evidence constraint. Figure 1 illustrates the problem and the method.

2 Related work

History-based counterfactual explainers. PRINCE [10] finds minimal sets of user actions that replace the top recommendation of a personalised-PageRank recommender over a heterogeneous graph. We do not run PRINCE at the scale of this study: it explains that recommender rather than

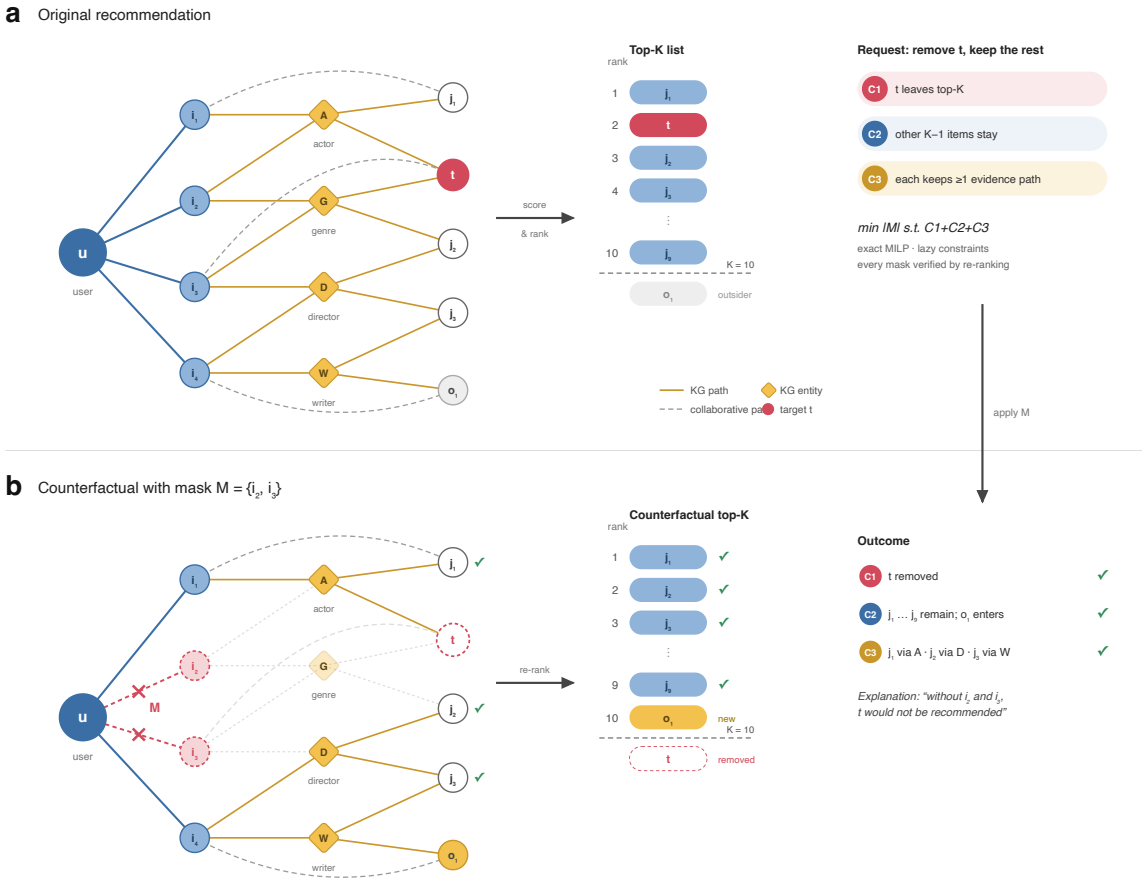


Figure 1: **EPIC-Rec removes one recommendation while keeping the rest of the top-K list and its evidence.** **a**, User u's history (i_1–i_4) reaches candidate items through knowledge-graph entities (diamonds; solid lines) and collaborative paths (dashed). Scores are sums of path contributions, giving a top-K list ($K = 10$) that contains the target t; o_1 is the best item outside the list. EPIC-Rec looks for the smallest set of history interactions M to mask such that t leaves the list (C1), the other $K-1$ items stay (C2), and each keeps at least one of its original evidence paths (C3). The problem is solved as a mixed-integer linear program for the margin problem P_ε on the permitted interactions $D_u(t)$, and every returned mask is checked by re-ranking. **b**, Masking $M = \{i_2, i_3\}$ removes their paths (faded). t leaves the list, j_1–j_9 remain, each supported by a surviving path (✓), and o_1 takes the free slot. The measured rates are in Figure 2.

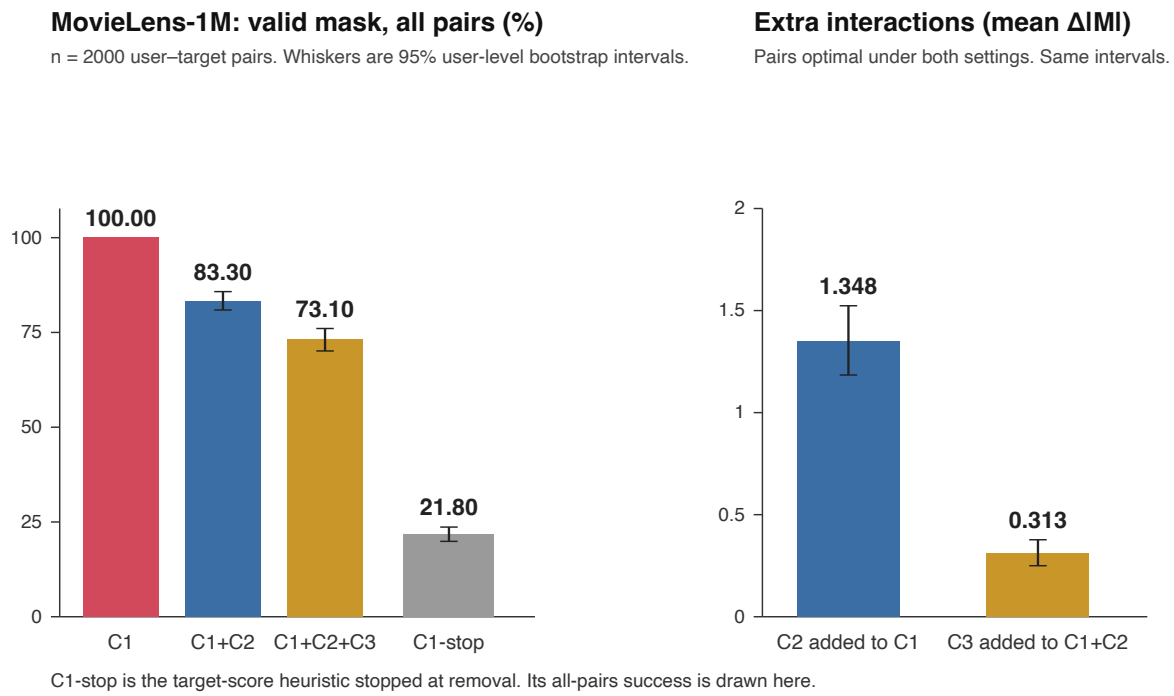


Figure 2: **MovieLens-1M rates on one denominator.** Both panels use all 2,000 pairs. Left: a mask exists for 100.00%, 83.30% and 73.10% of pairs under C1, C1 + C2 and C1 + C2 + C3. The grey bar is the target-score heuristic that stops at removal and, on those same pairs, happens to satisfy C1 + C2 + C3 (21.80%). Among only the edits that do remove the target, that share is 27.5%; that conditional rate is not drawn. Right: paired increase in minimum mask size when C2 is added to C1 (1.348 [1.185, 1.524]) and when C3 is then added (0.313 [0.250, 0.377]). Whiskers are 95% user-level bootstrap intervals. Last.fm is in Table 3.

the path-additive backbone, and no comparison under C2 or C3 is claimed. ACCENT [26] extends influence functions to pairs of items and removes the user's most influential interactions until the top recommendation is replaced. LXR [3, 11] trains an explainer network with counterfactual loss terms that scores the relevance of each piece of user data to a recommended item; the journal version adds per-explanation fine-tuning at inference time. SPINRec [4] computes path-integrated attributions for implicit-feedback data, sampling several plausible user profiles as baselines and keeping the most faithful integration path. These methods explain one recommended item; they do not constrain the rest of the list.

Graph explainers for recommendation. GREASE [6] trains a surrogate GNN around the user–item pair and optimises perturbations of its adjacency matrix to obtain factual and counterfactual explanations; it has been extended to top-k recommendation [7]. GNNUERS [18] perturbs the user–item graph to explain unfairness, and CF-GNNExplainer [17] removes edges to flip node predictions. CoSRec [32] learns a sparse retention mask over a user's interactions for a frozen LightGCN [13] with a ranking loss and a sparsity penalty, verifies the change by recomputation, and measures list stability. EPIC-Rec differs by enforcing membership as a hard constraint and by requiring evidence to survive.

Path-based explanations. Semantic knowledge has long been used to enrich usage-based recommendation [1], and knowledge-graph recommenders such as KGAT [30] and RippleNet [28] propagate preferences along entity paths. CERec [31] uses reinforced path reasoning to generate attribute-level counterfactuals over a knowledge graph, PEARLM [2] constrains path generation to paths that exist in the graph, and CountER [25] perturbs item aspects. None of them asks whether the paths that justify *other* recommendations survive a counterfactual edit.

Sequential and list-level evaluation. TRACE [16] produces training-free, ranking-aware bring-in and push-out counterfactuals for sequential recommenders from score queries. CFExpRec [21] and Beyond Top-1 [19, 20] show that evaluating explanations at top-1 gives inconsistent conclusions and propose top-K protocols. We follow this list-level view, but as a constraint on the explanation rather than only as an evaluation metric.

3 Problem formulation

Setting and ranking rule. User u has training history H_u. A fixed recommender scores every item of the candidate pool C_u, the catalogue minus H_u. For a mask $M \subseteq H_u$ let $s_M(j)$ be the score of j computed from $H_u \setminus M$. Items of C_u are ranked by s_M in descending order, *ties broken by item identifier in ascending order*; top-$K(M)$ is the first K items of this order. This one rule is used by the recommender, by the verifier and by every search method. $L_u = \text{top-}K(\emptyset)$ is the original list, $t \in L_u$ the target, $R = L_u \setminus \{t\}$ the retained items and $O = C_u \setminus L_u$ the outsiders.

Mask domain. The mask is restricted to $D_u(t) = \{i \in H_u : A_{it} > 0\}$, the interactions with at least one path to t (Section 4), so that an explanation names only interactions that support the target. On MovieLens-1M this restriction removes nothing: $D_u(t)$ covers 100.0% of the history, because collaborative paths connect every scored pair. On Last.fm it covers 70.3%.

Evidence unit: the individual path. The unit of evidence is one path $p = (i, \rho, x, j)$ from an interaction $i \in H_u$ through relation ρ and intermediate node x to item j. For a knowledge-graph

relation, x is an entity that i and j share through ρ ($i \xrightarrow{\rho} x \xleftarrow{\rho} j$); for the collaborative relation, x is another user who interacted with both i and j in the training data ($i \leftarrow x \rightarrow j$). The path's origin is $e(p) = i$; the path survives M when $e(p) \notin M$. Its contribution $c(p)$ is the amount it adds to $s(j)$ (Section 4); on our backbone it does not depend on what else is masked. Two paths that share an origin and a relation but pass through different entities or users are different pieces of evidence.

Definition 1 (Original evidence set)**.** *For a retained item $j \in R$ let $\tau_j = \max(\alpha \max_{p \to j} c(p),\ \phi)$, with $\alpha \in (0, 1]$ and the maximum taken over all paths from H_u to j. The floor $\phi \geq 0$ is a minimum contribution. The original evidence set of j is $P_j = \{p \to j : c(p) \geq \tau_j\}$. This study sets $\phi = 0$, so τ_j is the relative threshold. A path of contribution 0 is never evidence.*

The threshold is relative to the item's strongest single path, so the same α is comparable across users and items. A retained item with no path from H_u has an empty evidence set and cannot satisfy C3; such items make setting C infeasible for the pair, and we report how often this happens. On both main samples that count is 0 (MovieLens) and 0 (Last.fm). A surviving path is accepted as evidence only if disabling that single path lowers the recomputed score of j by a positive amount and by at least ϕ. That check is a property of the additive score in (1). It is not a claim that the path caused the user to like the item, or that a person would find the path a useful explanation.

Definition 2 (Problem P)**.** *Find $M \subseteq D_u(t)$ of minimum size such that*

C1 (removal)	$t \notin \text{top-}K(M)$,
C2 (membership)	$R \subseteq \text{top-}K(M)$,
C3 (evidence)	$\forall j \in R\ \exists p \in P_j :\ e(p) \notin M.$

Setting A asks for C1 only, setting B for C1 + C2, and setting C for C1 + C2 + C3. Setting A3, used only as an ablation, asks for C1 + C3 and drops the list constraint.

Retained items may reorder, and under C1+C2 exactly one outsider takes the freed position.

Margin variant P_ε. A MILP cannot express "ranked above, ties broken by identifier" without strict inequalities, so the solver works on a margin variant in which every ordering the constraints need holds with a gap of at least $\varepsilon = 10^{-6}$ in masked score:

$\mathbf{A}_\varepsilon$	$\big\lvert\{c \in C_u \setminus \{t\} : s_M(c) \geq s_M(t) + \varepsilon\}\big\rvert \geq K,$
$\mathbf{B}_\varepsilon$	$\exists r \in O :\ s_M(r) \geq s_M(t) + \varepsilon$ and, for every $j \in R$, $s_M(j) \geq s_M(t) + \varepsilon$ and $s_M(j) \geq s_M(c) + \varepsilon\ \ \forall c \in O \setminus \{r\},$
$\mathbf{C}_\varepsilon$	$\mathbf{B}_\varepsilon$ and C3.

Proposition 1. *Every mask feasible for P_ε is feasible for P in the same setting. Hence the P_ε optimum is an upper bound on the P optimum. The two problems can differ only when a mask needs an ordering with a gap in $[0, \varepsilon)$, in particular an exact tie resolved by item identifier.*

Proof sketch. Under A_ε, K items score strictly above t, so t is not in the first K whatever the tie rule. Under B_ε, each retained item is beaten at most by other retained items and by r, so it ranks within the first K; and t is beaten by the $K - 1$ retained items and by r. C3 does not involve the ranking. □

A proof of infeasibility for P_ε is therefore not a proof for P, and a P_ε optimum can exceed the P optimum. Infeasibility on $D_u(t)$ is likewise not infeasibility over the whole history: interactions

outside $D_u(t)$ are not decision variables. We measure the gap between P and P_ε by exhaustive search on every pair with $|D_u(t)| \le 14$ in a separate MovieLens sample (Section 6.2). That sample is evidence for those pairs, not a proof that the two problems agree in general. The main samples are not enumerated.

Scope of strict preservation. Related items—a sequel, a film by the same director—often share the target's evidence, and a user may expect them to disappear too. Strict $K-1$ preservation is therefore used in two roles: as a diagnostic that measures how entangled a recommendation is with the rest of the list, and as a single-item filter for users who endorse the remaining items.

4 Method

4.1 An inductive, additive path backbone

Study 1 uses a recommender whose score is an explicit sum over evidence paths:

$$s(u,j) = \beta \log(1+\mathrm{pop}_j) + \frac{1}{|H_u|} \sum_{i \in H_u} \sum_{\rho \in \mathcal{R}} \theta_\rho S_\rho[i,j], \tag{1}$$

where for a knowledge-graph relation ρ $S_\rho[i,j] = \sum_x 1/\deg_\rho(x)$ over the entities x that i and j share through ρ, and for the collaborative relation $S_{\mathrm{co}}[i,j] = \mathrm{co}(i,j)/\sqrt{\mathrm{pop}_i\,\mathrm{pop}_j}$, one term per co-interacting user. Each summand is one path p with contribution $c(p) = \theta_\rho/(\deg_\rho(x)|H_u|)$ or $c(p) = \theta_{\mathrm{co}}/(\sqrt{\mathrm{pop}_i \mathrm{pop}_j}\,|H_u|)$; the normalisations are frozen at their pre-mask values. Co-occurrence counts and popularity are computed on the training split. MovieLens keeps the 11 Satori film relations that survive the frequency filter; Last.fm keeps the 8 music relations released with KGCN [29]. Collaborative and trust-based signals remain central to recommendation [9]; our backbone represents the collaborative signal as explicit co-interaction paths, so it enters the evidence sets like any knowledge-graph relation. There is no user embedding, so masking removes all of M's signal and no retraining is needed. The weights $\theta_\rho \ge 0$ and $\beta \ge 0$ are learned with BPR [22] under non-negativity constraints on leave-one-out features: for a training triple (u,p,n) the model input is $H_u \setminus \{p\}$, and the co-occurrence and popularity statistics that contain u's own interaction with p are corrected accordingly.

Two consequences make the problem tractable. Masking interaction i subtracts a fixed row, $s_M(j) = s(j) - \sum_{i \in M} A_{ij}$ with $A_{ij} = W_{ij}/|H_u|$ and $W = \sum_\rho \theta_\rho S_\rho$, so every score is linear in the mask. And a surviving path's contribution does not depend on the mask, so with $P_{ij} = \max_{p:\, e(p)=i,\, p \to j} c(p)$ the largest single-path contribution from i to j, C3 holds for j if and only if some interaction in $Q_j = \{i : P_{ij} \ge \tau_j\}$ is unmasked.

4.2 Exact MILP for P_ε

Let $m_i \in \{0,1\}$ indicate $i \in M$ for $i \in D_u(t)$ and $z_r \in \{0,1\}$ select the replacement among a candidate set $\mathcal{O} \subseteq O$. Writing $\Delta_{ab}(m) = s(a) - s(b) - \sum_i m_i(A_{ia} - A_{ib})$, setting C is

$$\begin{aligned}
\min\ & \textstyle\sum_i m_i \\
\text{s.t.}\ & \textstyle\sum_{r \in \mathcal{O}} z_r = 1, \qquad \Delta_{rt}(m) \ge \varepsilon - B_{rt}(1-z_r) \ \ \forall r \in \mathcal{O}, && (2)\\
& \Delta_{jt}(m) \ge \varepsilon \ \ \forall j \in R, && (3)\\
& \Delta_{jc}(m) \ge \varepsilon - B_{jc} z_c \ \ \forall j \in R,\ c \in O \ \ (z_c \equiv 0 \text{ for } c \notin \mathcal{O}), && (4)\\
& \textstyle\sum_{i \in Q_j \cap D_u(t)} (1-m_i) \ge 1 \ \ \forall j \in R \text{ with } Q_j \subseteq D_u(t). && (5)
\end{aligned}$$

An interaction outside $D_u(t)$ has no mask variable: it is fixed unmasked. If Q_j meets the complement of $D_u(t)$, item j already keeps a qualifying origin that the program cannot delete, and (5) is omitted. Setting B drops (5); setting A3 keeps it and drops the list constraints; setting A replaces the ordering constraints by indicators y_c, one per candidate, and requires $\sum_c y_c \geq K$. Each big-M constant is the tightest valid value for its row, $B_{ab} = \max(0, \varepsilon - \min_m \Delta_{ab}(m))$. The score margin is $\varepsilon = 10^{-6}$. The study configuration sets HiGHS primal and MIP feasibility tolerances to 10^{-9}, three orders of magnitude smaller than ε. The verifier accepts a required gap down to $\varepsilon - 10^{-9}$.

Replacement pruning. Let $\mathrm{lb}(c) = s(c) - \sum_{i \in D_u(t)} A_{ic}$ be the lowest score c can reach. The replacement must satisfy $s(r) \geq \mathrm{lb}(t) + \varepsilon$, and we may further restrict it to $s(r) \geq \max_{c \in O} \mathrm{lb}(c)$. The reason: if a mask is feasible with replacement r', it is also feasible with b, the best outsider under the masked scores, because $s_M(b) \geq s_M(r') \geq s_M(t) + \varepsilon$ and every retained item already beats b (and hence r') by ε; and $s(b) \geq s_M(b) \geq s_M(c) \geq \mathrm{lb}(c)$ for every outsider c. The rule never excludes all candidates of a feasible instance; the regression tests compare the solver with pruning switched on and off.

Lazy constraints and outcomes. The threat constraints (4) are dense and numerous. We start with the 40 highest-scoring outsiders, solve, recompute the ranking under the returned mask, add every outsider that violates B_ε, and repeat. Every intermediate model is a relaxation, so an infeasible relaxation proves infeasibility of that relaxation, and a verified optimum of a relaxation is optimal for P_ε on $D_u(t)$ only when the lower bound matches the mask size. Each call passes HiGHS a 20-second limit. That limit is not a hard interrupt: HiGHS checks its clock periodically, so a returned solve can exceed 20 seconds. The recorded runtime is that wall-clock time. It includes building the model and the lazy rounds inside the call. It does not include the later cross-check, and it does not include the independent re-ranking, which is a separate pass. The solver is HiGHS [14] through `highspy` 1.15.1, one thread, relative MIP gap 0. The parent process kills a call that has not returned after 20 seconds plus a 40-second grace, and records the outcome as a timeout. No main-sample solve was killed (0 on MovieLens setting A, and likewise 0 for the other main settings). On MovieLens the median runtimes are 2.10, 0.97 and 0.80 seconds for C1, C1 + C2 and C1 + C2 + C3, and the maxima are 24.53, 24.32 and 24.39 seconds. The maximum above 20 seconds is the soft clock, not a second budget. Every instance ends in one of four outcomes: *optimal* (the verifier accepts the mask and the bound matches its size), *feasible* (the verifier accepts the mask, and minimality was not established within the budget), *infeasible* (infeasible for P_ε on $D_u(t)$), or *timeout*. A mask *exists* when the outcome is optimal or feasible. These runtimes are solver times on the study machine, not measured end-to-end latency of an interactive recommender.

A second HiGHS configuration, not a certificate. An optimality or infeasibility status is the status of one branch-and-bound run. It is not the same thing as the verifier accepting a mask, and it is not a mathematical certificate. On one MovieLens pair the study configuration (presolve off, feasibility tolerances 10^{-9}) returned a mask of size 4 as optimal, and a later run found a verified mask of size 3. We do not attribute that disagreement to a defect in HiGHS. The formulation, the presolve setting and the tolerances are all part of the computation. Every optimality or infeasibility claim in the tables was therefore re-solved for 60 seconds under a second configuration of the same solver: presolve on, primal tolerance 10^{-7}, MIP tolerance 10^{-6}. An optimum of size k is re-solved with $|M| \leq k - 1$. An infeasibility claim is re-solved with no size cap. If the second run is infeasible, the claim stands. If it returns a mask and the independent verifier accepts that mask, the smaller mask replaces the result and the new size is cross-checked the same way. Anything else is left

unresolved: a claimed optimum is reported as feasible, and a claimed infeasibility as a timeout. Two configurations of HiGHS are not two solvers, and a failed attempt to find a smaller mask is not a formal certificate of optimality. Optima of size 0 and infeasibility found by the structural pre-checks involve no search. The problem definitions are unchanged. On the MovieLens main sample the second configuration agreed with 7790 of the 8,000 exact results, replaced one optimum of size 4 by a verified optimum of size 3, and left 28 claimed optima unresolved. On Last.fm it agreed with all 8000 exact results. Where the mask domain has at most 14 interactions, the margin sample also enumerates P and P_ε directly (Section 6.2).

4.3 Exhaustive search, beam search and heuristics

Exhaustive search. The main samples are not enumerated. On the margin sample, when $|D_u(t)| \leq 14$, we enumerate masks in order of size and return the first feasible one, once under P and once under P_ε.

Beam search. C3 is anti-monotone under the additive backbone: once all of an item's evidence is masked, further masking cannot restore it. C1 and C2 are not: a retained item can leave the list after one deletion and return after a second. Our beam search (width 5, depth at most 15) searches under P_ε, prunes branches that violate C3, ranks the others by a margin surrogate, and checks the requested constraints at every node. The returned mask is minimal only within the beam.

Heuristics. The removal orders below are heuristics on our backbone, not re-implementations of any published explainer, and we make no claim that a published method reduces to any of them. Each adds at most 15 interactions and stops when the requested setting holds under P_ε. In settings with C3, an addition that would remove the last qualifying evidence of a retained item is skipped.

- *Violation* is dynamic: it adds the interaction that most reduces a hinge surrogate of the constraint violation.
- *Target-score attribution* removes interactions in descending order of A_{it}.
- *Ranking-margin attribution* is dynamic: at each step it removes the interaction with the largest $A_{it} - A_{ir}$, where r is the item currently at rank $K + 1$.
- *List-aware* removes in descending order of A_{it} minus the mean contribution of i to the retained items.
- *Popularity* and *random* are controls.

For the target-score order we also run a *C1-stop* mode, which adds interactions until C1 holds and then only reports whether C2 and C3 happen to hold. Two continuous relaxations are in-house baselines on this backbone, not published methods. Each interaction in $D_u(t)$ has a logit. The sigmoid relaxation sets the mask weight to $\sigma(w_i)$. The Gumbel relaxation uses a binary-concrete sample at temperature 0.5. The loss is a hinge on the ε-margins of the requested constraints, plus, when C3 is requested, a product over each retained item's qualifying origins, plus a sparsity penalty of 0.02 on the sum of the weights. Adam runs for 300 steps at learning rate 0.1 from the target-score attribution. Every 10 steps the weights are thresholded at 0.5, and the smallest mask that already satisfies the setting is kept. The final thresholded mask is then repaired by the same greedy additions and deletions used for the other heuristics, and the verifier accepts or rejects it.

Table 1: Surviving evidence for the four movies that stay in the worked example. Each row is one original path whose contribution is at least τ_j and whose origin is not masked. Entity names are not in the rehashed Satori release, so the intermediate node is a relation and a rehashed id, or "co-user".

Retained movie	τ_j	relation	origin	via	$c(p)$	survives
Godfather: Part II, The (1974)	0.086315	actor	Godfather, The (1972)	rehashed-entity-2339	0.172631	yes
Shawshank Redemption, The (1994)	0.129473	actor	Insider, The (1999)	rehashed-entity-3141	0.258946	yes
Star Wars: Episode IV - A New Hope (1977)	0.064737	actor	Bridge on the River Kwai, The (1957)	rehashed-entity-2349	0.073985	yes
Pulp Fiction (1994)	0.016184	actor	Sixth Sense, The (1999)	rehashed-entity-1401	0.032368	yes

Verification. Every mask that any method returns is re-ranked from the model, not from the solver's matrix, and is accepted only when the requested constraints hold under both P and P_ε. A mask that fails is reported as a verification failure and is never counted as a success.

Regression tests. The test suite compares the MILP, with replacement pruning on and off, to exhaustive search of P_ε on synthetic instances with ties and on small instances from the trained model. Those comparisons are not part of the reported samples.

4.4 Worked example: removing one recommendation while preserving the others

The example is one MovieLens pair from the stored $K = 5$ sweep (exact method, $\alpha = 0.5$, $\varepsilon = 10^{-6}$, mask domain $D_u(t)$). It was chosen because setting C is recorded as optimal with a mask of size 1, the history has 12 movies, the mask domain has 12 interactions, and the cross-check status is confirmed. The target is a widely known title. It illustrates the procedure. It is not an estimate of average mask size or feasibility.

User 4739 is recommended, in order, 1. *Godfather: Part II, The (1974)* (11.3328), 2. *Shawshank Redemption, The (1994)* (11.0729), 3. *Star Wars: Episode IV - A New Hope (1977)* (10.9434), 4. *Star Wars: Episode VI - Return of the Jedi (1983)* (10.2361), 5. *Pulp Fiction (1994)* (9.6114). The target is *Star Wars: Episode VI - Return of the Jedi (1983)*, originally rank 4, score 10.2361. The three history movies that contribute most to its score are *Star Wars: Episode V - The Empire Strikes Back (1980)* (2.6780), *Star Wars: Episode I - The Phantom Menace (1999)* (1.3932), *Sixth Sense, The (1999)* (0.8205).

The masked interaction is *Star Wars: Episode V - The Empire Strikes Back (1980)*, which is among those three contributors. Its contribution is removed and the others stay.

Under C1+C2+C3 the solver returns M of size 1: Star Wars: Episode V - The Empire Strikes Back (1980). The study configuration reports status optimal in 0.1137 seconds. The independent verifier accepts the mask under both P and P_ε (C1 = True, C2 = True, C3 = True). The stored $K = 5$ record has status optimal, size 1, and the masks match. Enumeration of every subset of $D_u(t)$ (12 interactions) finds the same minimum size under P (1) and under P_ε (1). No smaller permitted mask satisfies C1+C2+C3 under either formulation.

After the mask the top five are 1. *Godfather: Part II, The (1974)* (10.3997), 2. *Shawshank Redemption, The (1994)* (10.1131), 3. *Pulp Fiction (1994)* (8.6997), 4. *Star Wars: Episode IV - A New Hope (1977)* (8.6464), 5. *Schindler's List (1993)* (8.5450). *Star Wars: Episode VI - Return of the Jedi (1983)* is not among them. The other four original movies remain. The replacement is *Schindler's List (1993)* (rank 5, score 8.5450). Kendall's τ_a on the four retained movies, before against after, is 0.667.

Masking Star Wars: Episode V - The Empire Strikes Back (1980) removes *Star Wars: Episode VI - Return of the Jedi (1983)* from the top-five recommendations. The remaining four original movies stay in the list, and each retains at least one original evidence path meeting its contribution

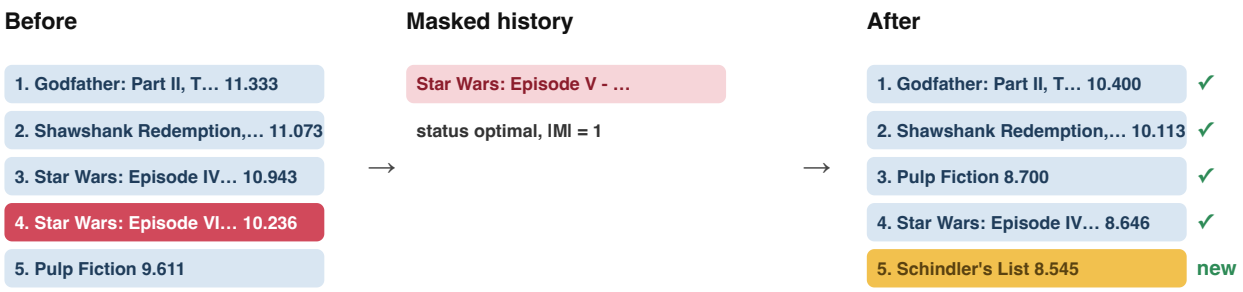


Figure 3: Worked example, $K = 5$. Left: the original list; the target is red. Centre: the masked history interaction. Right: the list after re-ranking. Blue rows are retained movies, the gold row is the replacement, and a tick marks a retained movie that keeps a qualifying path. Scores are the path-additive scores.

threshold. The edit is a change to the inputs of the scorer, followed by re-ranking. It is not a change to the interface, and this example is not evidence about privacy, conversion or trust.

On this pair the target-score heuristic that stops at removal masks Star Wars: Episode V - The Empire Strikes Back (1980). The verifier reports C1 = True, C2 = True, C3 = True. This particular removal happens to satisfy C2 and C3 as well, so it is not an example of collateral damage. The main-sample rate, not this pair, is the evidence that removal-only edits usually do not.

5 Experimental setup

Data. MovieLens-1M [12] ratings of at least 4 are positives. Movies are linked to the Microsoft Satori knowledge graph through the public RippleNet release [28]. Last.fm artist plays are the HetRec 2011 release [5], linked to Satori through the public KGCN release [29]; every play count of at least 1 is a positive. We keep items with a knowledge-graph link, then a core of 10 users and 5 items on MovieLens and of 5 users and 3 items on Last.fm, and we drop entities shared by only one remaining item. Relations with fewer than 200 links on MovieLens, or 20 on Last.fm, are dropped. This leaves 5,646 users and 1,979 movies on MovieLens, all of them still linked, and 1,716 users and 1,286 artists on Last.fm, of which 1,010 keep a link after the relation filter. Each user's positives are shuffled and split 70/10/20 into train, validation and test. Co-occurrence, popularity and the path weights are fit on the training split only. Validation chooses the baseline hyperparameters. Explanations are computed on the training history, which is the input of the deployed model.

Backbone accuracy. Table 2 reports Recall@20 and NDCG@20 on the test split, ranked with the same tie rule as the solver. The backbone is competitive with item-based collaborative filtering and with BPR matrix factorisation on both datasets, so the explanations concern a reasonable recommender rather than a toy one. The learned weight on collaborative paths is $\theta_{\text{co}} = 31.79$ on MovieLens and 42.78 on Last.fm. The popularity weight is $\beta = 0.00$ on MovieLens and 0.48 on Last.fm. The knowledge-graph-only row is a separately trained model with the collaborative relation removed.

Protocol. On each dataset we draw 50 users from each quartile of training-history length (200 users, seed 1) and take every item of each user's top-10 list as a target, giving 2,000 pairs. The main

Table 2: Test-set top-20 accuracy (%). MovieLens: 5,646 users. Last.fm: 1,716 users. ItemKNN uses cosine similarity; validation selected the full neighbourhood over neighbourhood sizes all, 50 and 200. BPR-MF has 64 factors and stopped after 10 epochs on the validation split.

	MovieLens-1M		Last.fm	
Model	Recall@20	NDCG@20	Recall@20	NDCG@20
Popularity	20.7	17.6	22.0	10.5
ItemKNN	28.2	25.8	39.5	23.3
BPR-MF	26.7	22.8	31.5	17.8
Backbone, KG paths only	21.8	19.6	22.4	11.8
Backbone, KG and collaborative paths	29.1	26.5	42.2	25.3

Table 3: Solver outcomes after the second HiGHS configuration (%). Each row is 2,000 pairs from 200 users, $K = 10$, $\alpha = 0.5$, $\varepsilon = 10^{-6}$. A mask exists when the outcome is optimal or feasible. The feasible share on MovieLens includes the 28 claimed optima that the second HiGHS configuration did not confirm. The interval resamples users.

Dataset	Setting	Optimal	Feasible	Infeasible	Timeout	Exists [95% CI]
MovieLens-1M	C1	95.30	4.70	0.00	0.00	100.00 [100.00, 100.00]
MovieLens-1M	C1 + C3	83.85	0.75	15.40	0.00	84.60 [82.29, 86.74]
MovieLens-1M	C1 + C2	80.35	2.95	16.55	0.15	83.30 [80.92, 85.75]
MovieLens-1M	C1 + C2 + C3	71.20	1.90	26.90	0.00	73.10 [70.11, 76.05]
Last.fm	C1	95.70	0.00	4.30	0.00	95.70 [94.75, 96.60]
Last.fm	C1 + C3	82.85	0.00	17.15	0.00	82.85 [80.80, 84.75]
Last.fm	C1 + C2	30.05	0.00	69.95	0.00	30.05 [28.05, 31.90]
Last.fm	C1 + C2 + C3	28.05	0.00	71.95	0.00	28.05 [26.15, 30.00]

configuration is $K = 10$, $\alpha = 0.5$, $\phi = 0$ and $\varepsilon = 10^{-6}$. Three further MovieLens samples, 25 users per quartile, vary one factor each: the evidence threshold $\alpha \in \{0.25, 0.5, 0.75, 1\}$ under C1 + C2 + C3; the list size $K \in \{5, 10, 20\}$; and the margin $\varepsilon \in \{10^{-8}, 10^{-6}, 10^{-4}\}$, with exhaustive search wherever $|D_u(t)| \leq 14$. These samples are not the main sample. The exact solver has a budget of 20 seconds. The second HiGHS configuration uses 60 seconds per re-solve.

Statistics. Mask-size costs are paired. A difference is computed only on the pairs that are optimal in both compared settings, because the hardest pairs drop out as constraints tighten. Each table states how many pairs and users lie behind a difference. All 95% intervals are percentile bootstrap intervals (2,000 replicates) that resample users within each history quartile and keep every target of a resampled user together.

6 Results

6.1 How often is surgical removal possible?

Table 3 is the outcome of every main-sample pair after the second HiGHS configuration. On MovieLens, removing the target alone is always possible: a mask exists for 100.00% of pairs (95.30% whose optimum the second configuration agreed with, 4.70% with a verified mask whose minimality it did not settle). Keeping the other nine items lowers this to 83.30% [80.92, 85.75], and keeping

Table 4: Paired cost of a constraint: difference in minimum mask size, on the pairs that are optimal in both settings. The interval resamples users.

Dataset	Comparison	Pairs (users)	Mean [95% CI]	Median	Share > 0
MovieLens-1M	C1+C2 − C1	1535 (198)	1.348 [1.185, 1.524]	0	33.7%
MovieLens-1M	C1+C2+C3 − C1+C2	1401 (198)	0.313 [0.250, 0.377]	0	15.1%
MovieLens-1M	C1+C3 − C1	1598 (200)	0.397 [0.325, 0.479]	0	18.9%
Last.fm	C1+C2 − C1	601 (182)	0.271 [0.212, 0.334]	0	20.0%
Last.fm	C1+C2+C3 − C1+C2	561 (176)	0.007 [0.002, 0.014]	0	0.7%
Last.fm	C1+C3 − C1	1657 (200)	0.020 [0.012, 0.029]	0	1.8%

their evidence as well lowers it to 73.10% [70.11, 76.05]. The pairs without a mask are reported infeasible for P_ε on $D_u(t)$ (16.55% under C1 + C2, 26.90% under C1 + C2 + C3) except for 0.15% of pairs under C1 + C2, which reached neither a mask nor an infeasibility status. That status is not an infeasibility proof for P, and not for masking outside $D_u(t)$. On Last.fm, removal itself fails for 4.30% of pairs. Keeping the list is the binding constraint: existence falls to 30.05% [28.05, 31.90], and requiring the evidence as well leaves 28.05% [26.15, 30.00]. Requiring the evidence without the list constraint, C1 + C3, leaves existence at 84.60% on MovieLens and 82.85% on Last.fm, so the evidence constraint is affordable once the target is removed and the list constraint is not. No main-sample pair has a retained item with an empty evidence set. No solve was killed. Median solve times are 2.10, 0.97 and 0.80 seconds for the three MovieLens settings; the longest MovieLens solve took 24.53 seconds.

Of the Last.fm pairs with no C1 + C2 + C3 mask, 91.2% are already impossible under C1 + C2. On MovieLens that share is 61.5%; evidence entanglement accounts for 18.2% and a joint conflict, in which C1 + C2 and C1 + C3 are each feasible but not together, for 19.9%. On Last.fm those two causes are 0.4% and 2.4%.

Table 4 is the cost on the pairs where both minima are confirmed. On MovieLens, keeping the list costs 1.348 [1.185, 1.524] extra interactions on average (1535 pairs, 198 users), and keeping the evidence costs another 0.313 [0.250, 0.377] (1401 pairs). Both medians are 0, and only 33.7% and 15.1% of the pairs need any extra interaction, so the averages are carried by a minority of entangled pairs. On Last.fm the same two costs are 0.271 [0.212, 0.334] and 0.007 [0.002, 0.014]. The evidence cost is positive but tiny: once a list-preserving mask exists, one more interaction is rarely required. No pair needs a smaller mask under a stricter setting. Under the minimum masks the retained items reorder only moderately. Mean Kendall's τ between their original and new order is 0.73 and 0.77 on MovieLens and 0.71 and 0.71 on Last.fm.

Rank matters most. Figure 4 and Table 5 show the same rank effect on both datasets. A list-preserving mask exists for 54.0% of MovieLens rank-1 targets and for 98.5% of rank-10 targets (C1 + C2 + C3: 36.0% and 98.5%). On Last.fm the rank-1 rates are 2.5% and 1.5%, against 63.5% and 60.5% at rank 10. A borderline recommendation can usually be removed without side effects. The strongest recommendation usually cannot, and on Last.fm it almost never can.

Short histories are more entangled. Table 6 stratifies by history length. The shortest quartile has the lowest existence under C1 + C2 and under C1 + C2 + C3: 57.2% and 42.8% on MovieLens, against 99.6% and 96.4% in the longest quartile, and 11.8% and 9.4% against 45.2% and 43.6% on Last.fm. With few interactions, each one supports a large part of the list.

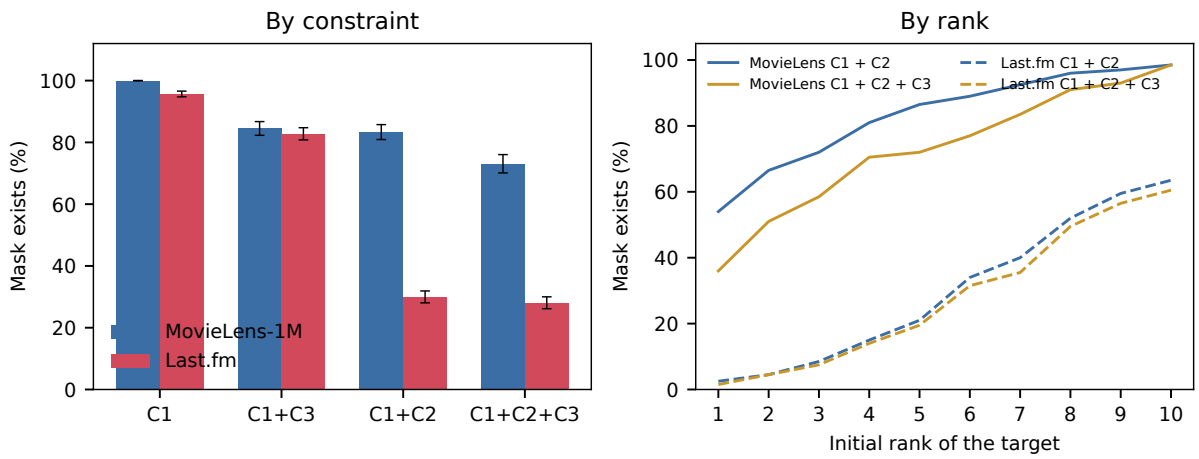


Figure 4: Mask existence after the second HiGHS configuration. Left: by constraint, with the 95% user-level bootstrap interval. Right: by the target's initial rank, for C1 + C2 (blue) and C1 + C2 + C3 (gold); solid lines are MovieLens-1M and dashed lines are Last.fm. Each point is 200 pairs.

Table 5: Mask existence (%) by the target's initial rank. Each cell is 200 pairs.

Rank	ML C1+C2	ML C1+C2+C3	LF C1+C2	LF C1+C2+C3
1	54.0	36.0	2.5	1.5
2	66.5	51.0	4.5	4.5
3	72.0	58.5	8.5	7.5
4	81.0	70.5	15.0	14.0
5	86.5	72.0	21.0	19.5
6	89.0	77.0	34.0	31.5
7	92.5	83.5	40.0	35.5
8	96.0	91.0	52.0	49.5
9	97.0	93.0	59.5	56.5
10	98.5	98.5	63.5	60.5

6.2 Original problem versus margin variant

Table 7 compares P with P_ε at three margins. The enumerable pairs are 190 pairs from 19 of the 100 sampled users. Feasibility and the optimum never differ, and the MILP agrees with the P_ε enumeration on every decided pair–setting combination. The two problems can differ when a mask needs a gap smaller than ε or an exact tie. On this sample, at these margins, they did not. On the pairs that are too large to enumerate, the indirect check is the one in Table 8: across every stored comparison of an approximate method with the reported optimum, 0 masks were smaller than that optimum and 0 approximate masks succeeded on a pair the MILP reported as infeasible.

6.3 Do constraint-unaware edits preserve the list?

The target-score order that stops at removal reaches C1 for 79.15% of MovieLens pairs and 94.65% of Last.fm pairs. Among those edits, 35.8% and 13.0% keep the other nine items, and 27.5% and 12.4% keep the items and their evidence. Stability is not a by-product of a minimal removal: it

Table 6: Mask existence (%) by training-history quartile, shortest to longest. Each cell is 500 pairs.

History quartile	ML C1+C2	ML C1+C2+C3	LF C1+C2	LF C1+C2+C3
shortest	57.2	42.8	11.8	9.4
second	82.6	66.4	27.2	24.8
third	93.8	86.8	36.0	34.4
longest	99.6	96.4	45.2	43.6

Table 7: P versus P_ε on the MovieLens margin sample, for every pair with $|D_u(t)| \leq 14$. Both problems are solved by exhaustive search. The agreement column sums the three settings, so the denominator is three times the number of pairs. The main samples were not enumerated.

ε	Pairs (users)	Status or optimum differs	MILP agrees
10^{-8}	190 (19)	0	570/570
10^{-6}	190 (19)	0	570/570
10^{-4}	190 (19)	0	570/570

has to be asked for. When the same orders are required to satisfy the constraints, beam search and the violation greedy come closest to the exact solver. Beam search returns a verified mask on 63.45% of MovieLens pairs and on 28.05% of Last.fm pairs, which is every Last.fm pair for which a mask exists, and the mask has the reported size for 95.90% and 100.00% of its successes on pairs the exact method reported as optimal. The violation greedy is close (62.05% and 27.15% success). The repaired continuous masks succeed more often than beam search on MovieLens and less often at the reported size (Table 8). Popularity and random orders rarely satisfy the constraints. No approximate mask in the stored comparisons is smaller than a reported optimum (0), and none succeeds on a pair the exact method reported as infeasible (0).

6.4 Sensitivity

Table 9 raises the evidence threshold. Mask existence under C1 + C2 + C3 falls from 79.8% at $\alpha = 0.25$ to 69.9% at $\alpha = 1$. The mean size on the pairs that remain optimal does not grow, because the pairs that become infeasible leave the average. Table 10 varies the list. A list-preserving mask exists for 93.20% of pairs at $K = 5$ and 70.95% at $K = 20$ (C1 + C2 + C3: 88.00% and 54.70%). Removal alone is solved for every pair at every K. At $K = 20$, 1.75% of the C1 + C2 pairs time out. Longer lists are harder to preserve because more items share the target's support.

7 Discussion and limitations

What the numbers mean. On MovieLens, most recommendations, and almost all borderline ones, have a small edit that removes the item and leaves the rest of the list and its evidence intact. The strongest recommendations, and the users with the shortest histories, often do not. On Last.fm the list constraint binds much earlier: a list-preserving edit exists for about a third of pairs, and for the top-ranked item almost never. In both datasets the failures are mostly reported infeasible for P_ε on $D_u(t)$, not merely unsolved. The formulation turns that entanglement into a measured quantity. One MovieLens optimum of size 4 was replaced by a verified optimum of size 3, and 28 claimed MovieLens optima remain feasible rather than optimal because a second configuration of

Table 8: Approximate methods under C1 + C2 + C3. Success is the share of all 2,000 pairs for which the verifier accepts the mask. Exact size is the share of those accepted masks, among pairs with a reported optimum, that have that optimum's size. For the stop-at-C1 row, success is the share of pairs whose removal-only mask happens to satisfy C1 + C2 + C3.

	MovieLens-1M		Last.fm	
Method	success	exact size	success	exact size
beam search	63.45	95.90	28.05	100.00
violation greedy	62.05	90.72	27.15	98.89
continuous, sigmoid	71.30	68.52	26.60	97.18
continuous, Gumbel	71.35	63.50	26.30	95.25
ranking-margin order	45.40	75.96	14.35	91.64
list-aware order	44.20	72.18	18.75	93.33
target-score order	32.10	63.56	12.85	95.72
target-score, stop at C1	21.80	76.17	11.75	97.45
popularity order	15.10	37.25	6.45	75.97
random order	15.10	26.49	8.15	77.30

Table 9: Evidence threshold α on a separate MovieLens sample (100 users, 1,000 pairs, C1 + C2 + C3, $K = 10$). Outcomes are percentages. The mean size is over the pairs whose optimum was confirmed, so pairs that become infeasible drop out of it.

α	Exists	Optimal	Infeasible	Mean $\lvert M\rvert$ on optima
0.25	79.8	76.0	20.2	7.19
0.50	76.2	73.0	23.8	6.99
0.75	72.9	70.8	27.1	6.99
1.00	69.9	68.3	30.1	6.92

HiGHS did not confirm them. That cross-check is not a certificate.

Why Last.fm is harder. Full-constraint existence is 73.10% on MovieLens and 28.05% on Last.fm. The drop is already present under C1 + C2: 83.30% against 30.05%. Adding C3 changes Last.fm only from 30.05% to 28.05%. On the Last.fm pairs where C1 + C2 + C3 is infeasible, the recorded cause is list entanglement for 91.2% of those pairs. Evidence entanglement accounts for 0.4% and a joint conflict for 2.4%. Two stored differences travel with this gap, and this paper does not separate them experimentally. The learned popularity weight is $\beta = 0.00$ on MovieLens and $\beta = 0.48$ on Last.fm; the weights were fit separately, so this is not a controlled ablation with the other weights held fixed. The permitted mask domain $D_u(t)$ covers 100.0% of training-history interactions on the MovieLens main pairs and 70.3% on Last.fm. No main-sample pair on either dataset has an empty evidence set (0 and 0). A retrained popularity ablation, and a run that expands the mask domain to the whole history, were not done. The association is that Last.fm fails the list constraint, not that

Table 10: List size K on a separate MovieLens sample (100 users, every top-K item). Outcomes are percentages. The $K = 10$ row is not the main sample.

	C1		C1 + C2		C1 + C2 + C3	
K	exists	timeout	exists	timeout	exists	timeout
5	100.00	0.00	93.20	0.00	88.00	0.00
10	100.00	0.00	84.70	0.20	74.90	0.10
20	100.00	0.00	70.95	1.75	54.70	0.00

its evidence paths are uniquely entangled.

Limitations. (i) The backbone is additive by design, so that path contributions are exact and C3 can be checked in closed form. Non-additive recommenders need a single-path-disabling definition of contribution and an approximate solver; this is study 2. (ii) The exact solver works on P_ε with $M \subseteq D_u(t)$. Agreement with P was checked by enumeration on the 190 enumerable pairs of the margin sample, from 19 users, mostly short histories. That is not a general equivalence. Infeasibility for P_ε, or outside $D_u(t)$, is not infeasibility for P on the whole history. (iii) The sensitivity samples have 100 users and are separate from the main 200. (iv) The heuristics, the beam and the continuous masks are our own procedures on this backbone. They are not GREASE, LXR, ACCENT, PRINCE, TRACE or the counterfactual support-set method of Zhang et al. Those methods were not run. The blocker is concrete. GREASE trains a surrogate GNN and perturbs its adjacency; ACCENT uses influence functions on a different model; LXR trains an explainer network; PRINCE explains personalised PageRank; Zhang et al. learn a retention mask for a frozen LightGCN. None of their official implementations is a solver for P_ε on this path-additive backbone, and adapting them would be a new experiment. No number in this paper is such an experiment. Zhang et al. is cited from the authors' preprint; the venue and DOI were not confirmable from the publisher, and the acronym CoSRec also names an unrelated SIGIR 2025 dataset, so the paper is cited by title. (v) The knowledge graphs are the Satori linkages released with RippleNet and KGCN, not KB4Rec or Freebase. Entity names are not in the rehashed release. (vi) The literature check was focused rather than systematic. We do not claim to be the first method of this kind. (vii) The backbone is static. User preferences drift over time [23, 24]; with time-dependent scores the score row of an interaction would change between requests, and extending evidence-preserving counterfactuals to such dynamic recommenders is future work.

8 Conclusion

EPIC-Rec asks for counterfactual explanations that remove one recommendation without collateral damage to the rest of the list or to its evidence. On this additive path recommender the margin problem on $D_u(t)$ has a mixed-integer formulation, and returned masks are re-ranked independently. On MovieLens a feasible mask is usual, except for the items the user is most strongly recommended. On Last.fm, keeping the rest of the list is possible for only a minority of recommendations. The formulation gives a test bed for explainers that claim list-level faithfulness. It does not by itself establish that claim for other backbones.

Code and data availability. The implementation is at `https://github.com/MehrdadJalali-AI/EPIC-Rec`. The commit that contains this revision is recorded in `manuscript/REVISION`.

Aggregated tables in `results/report/tables` and the compact per-instance exact statuses in `results/published` are the machine-readable results behind the tables and Figure 2. `scripts/manuscript_from_report.py` writes `numbers.tex` and the result tables from those aggregates. `scripts/fig1_from_results.py` draws the charts in Figure 1 from the same files. The worked example is regenerated by `scripts/export_worked_example.py`. The prose is not generated automatically. The full `records.jsonl` files are produced by the runner and are not stored in git because each main file is about 80 MB; the commands that regenerate them are in `docs/reproducibility.md`. Raw datasets and credentials are not redistributed. HiGHS is used through `highspy` 1.15.1.

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